BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6483

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If rate equation of reaction $2A + B \rightarrow$ Product, is rate = $K[A]^2[B]$ and A is present in large excess, then order of reaction is:

(A) 1
(B) 2
(C) 3
(D) 4

Q2. The oxidation state of 'Mn' in $KMnO_4$ is:

(A) + 7
(B) + 6
(C) + 2
(D) + 5

Q3. 18g glucose dissolved in 90gm of H_2O has relative lowering of vapour pressure equal to:

(A) 18/90
(B) 1/6
(C) 10/51
(D) 1/51

Q4. pH of human blood is:

(A) 7.35
(B) 6.35
(C) 5.35
(D) 4.35

Q5. For a given process, the heat changes at constant pressure (q_p) and at constant volume (q_v) are related to each other as:

(A) $q_p = q_v$ (B) $q_p < q_v$

(C) $q_p > q_v$ (D) $q_p = q_v/2$

Q6. Which of the hydrogen halides has the highest percentage ionic character?

(A) HCl (B) HBr

(C) HF (D) HI

Q7. Ionization energy for $Mg \rightarrow Mg^+ + 1e$ has ΔH = ?

(A) 738 KJ mol^{-1} (B) 238 KJ mol^{-1}

(C) 448 KJ mol^{-1} (D) 138 KJ mol^{-1}

Q8. Splitting of spectral lines when atoms are subjected to strong electrical field is called:

(A) Zeeman effect (B) Stark effect

(C) Photoelectric effect (D) Compton effect

Q9. De-Brogli equation is represented as:

(A) $h = \lambda/mv$ (B) $m = h/\lambda v$

(C) $m = h/\lambda$ (D) $\lambda = h/mv$

Q10. The molecules of CO_2 in dry ice form the:

(A) Ionic crystals (B) Covalent crystals

(C) Molecular crystals (D) Metallic crystals

Q11. Density of ice is minimum at 4°C due to:

(A) Empty spaces in structure of ice (B) Tetrahedral shape of crystal of ice

(C) Large bond lengths (D) Large bond angles

Q12. The temperature of a natural plasma is about:

(A) 20000°C (B) 1000°C

(C) 5000°C (D) 10000°C

Q13. The deviation of a gas from ideal behaviour is maximum at:

(A) 0°C and 2.0 atm (B) -10°C and 5 atm

(C) 100°C and 2 atm (D) -10°C and 2 atm

Q14. The technique of chromatography is useful in organic synthesis for:

(A) Separation (B) Isolation

(C) Purification (D) All these

Q15. Separating funnel is used in the technique of analysis:

(A) Crystallization
(B) Filtration
(C) Solvent extraction
(D) Sublimation

Q16. Nickel has number of isotopes:

(A) 3
(B) 5
(C) 7
(D) 2

Q17. The number of moles of CO_2, which contain 8.0g of oxygen:

(A) 0.25
(B) 0.50
(C) 1.0
(D) 1.50

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6483

Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) Magnesium atom is twice heavier than that of carbon atom. Justify it.

(ii) Many chemical reactions taking place in our surrounding involve the limiting reactants. Give examples.

(iii) Molecular formula is multiple of empirical formula. Give an example.

(iv) How is chromatography classified on the basis of stationary phases?

(v) Define sublimation. Give two examples.

(vi) Write names of four steps of crystallization.

(vii) Calculate the value of general gas constant (R) using S.I. units of pressure and volume.

(viii) What is Joule-Thomson effect?

(ix) Write quantitative definition of Charles's law.

(x) State Le-Chatelier's principle.

(xi) How does equilibrium constant tell about direction of reaction?

(xii) What is the effect of common ion on solubility? Give an example.

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) Amorphous solid like glass is also super cooled liquid. Why?

(ii) Cleavage of crystal is itself anisotropic behaviour. Justify it.

(iii) Water and ethanol can mix easily in all proportions. Give reason.

(iv) In a cold winter the fish in garden ponds owe their lives to H-bonding. Explain.

(v) Define Hunds rule with an example.

(vi) Give out two defects of Rutherford Model of an atom.

(vii) Differentiate between Zeeman and Stark effect.

(viii) Define continuous spectrum with an example.

(ix) Why some of properties are called colligative?

(x) What are the conditions to obey colligative properties?

(xi) Define half life time (period) with an example.

(xii) How the surface area affect the rate of reaction?

Q4. (Marks 12) Write short answers of any SIX parts.

(i) Why atomic radius cannot be determined precisely?

(ii) How ionization energy changes in periodic table?

(iii) What is coordinate covalent bond? Give one example.

(iv) Why bond order of Helium molecule is zero?

(v) Why enthalpy of neutralization is called enthalpy of formation of H_2O?

(vi) Define heat capacity of a body. Give its mathematical expressions.

(vii) What is enthalpy of reaction? Give example.

(viii) What is oxidation number? Give example.

(ix) Write the product obtained during electrolysis of $PbBr_2$.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5. (Marks 02,02) (a)Describe combustion analysis. Also write formula to calculate percentage of carbon, hydrogen and oxygen.

(b) State Mosley's law. What is its importance?

Q6. (Marks 04) (a)250cm^3 of hydrogen gas is cooled from 127°C to -27°C keeping the pressure constant. Calculate the new volume of the gas at low temperature.

(b) Explain the construction and working of fuel cells.

Q7. (Marks 1,3) (a)Give the assumptions and postulates of VSEPR theory.

(b) Define and explain Hess's law of constant heat summation with an example.

Q8. (Marks 3,1) (a)Write the structure of ice. Why ice floats on water?

(b) The solubility product of Ag_2CrO_4 is 2.6×10^{-2} at 25°C. Calculate the solubility of the compound.

Q9. (Marks 04) (a)How lowering of vapour pressure as colligative property is used to find out molecular mass of solute?

(b) Explain any four characteristics of a catalyst.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6486

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Ionic solids are characterized by:

(A) Low melting point
(B) Good conductors in solid state
(C) High vapour pressure
(D) Solubility in polar solvents

Q2. London dispersion forces are present among the:

(A) Molecules of liquid water
(B) Molecule of hydrogen chloride gas
(C) Molecule of solid iodine
(D) All these

Q3. Value of R at STP:

(A) 8.21 dm^3 atm k^{-1} mol^{-1}
(B) 0.0821 dm^3 atm k^{-1} mol^{-1}
(C) 0.00821 dm^3 atm k^{-1} mol^{-1}
(D) 0.000821 dm^3 atm k^{-1} mol^{-1}

Q4. Gases deviate from ideal behaviour at high pressure because:

(A) At high pressure, the gas molecule move in one direction only
(B) At high pressure, the gas molecules move in all direction
(C) At high pressure, there are significant attractive forces
(D) All these

Q5. Coloured impurities appear during crystallization are removed by boiling the substance in the solvent with:

(A) Silica gel
(B) Benzoic acid
(C) Powdered animal charcoal
(D) $CaCl_2$

Q6. A technique in which a solute distribute itself in stationary phase and mobile phase is called:

(A) Sublimation
(B) Solvent extraction
(C) Chromatography
(D) None of these

Q7. Many elements have fractional atomic masses. This is because:

(A) Mass of the atom is itself fractional
(B) Atomic mass are average masses of isobars
(C) Atomic masses are average masses of isotopes
(D) Atomic masses are average masses of isotopes proportional to their relative abundance

Q8. The volume occupied by 1.4g of N_2 at S.T.P. is:

(A) 2.24 dm^3
(B) 22.4 dm^3
(C) 1.12 dm^3
(D) 112 cm^3

Q9. The catalytic activity of enzyme is greatly enhanced by the presence of:

(A) Inhibitors
(B) Coenzymes
(C) Activators
(D) Coenzymes & activators

Q10. Oxidation number of 'Mn' in $KMnO_4$ is:

(A) 3
(B) 5
(C) 7
(D) 9

Q11. 18gram glucose is dissolved in 90gram of water. The relative lowering of vapour pressure equal to:

(A) 1/5
(B) 5.1
(C) 1/51
(D) 6

Q12. pH of 10^{-4} mol dm^{-3} of HCl is:

(A) 1
(B) 2
(C) 3
(D) 4

Q13. For the reaction $NaOH + HCl \rightarrow NaCl + H_2O$ the change in enthalpy is:

(A) Heat of reaction
(B) Heat of formation
(C) Heat of neutralization
(D) Heat of combustion

Q14. Bond order for He_2 is:

(A) 0
(B) 1
(C) 2
(D) 3

Q15. Ethyne molecule have:

(A) Three π bonds between carbon atom

(B) Three σ bonds between carbon atom

(C) One σ and two π bonds between carbon atom

(D) One π and two σ bonds between carbon atom

Q16. Quantum number value for 2p orbitals are:

(A) $n = 2, \ell = 1$

(B) $n = 1, \ell = 2$

(C) $n = 1, \ell = 0$

(D) $n = 2, \ell = 0$

Q17. In the ground state of an atom, the electron is present:

(A) In the nucleus

(B) In the second shell

(C) Nearest to the nucleus

(D) Farthest from the nucleus

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6486

Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) Define isotopes. Write isotopes of carbon.

(ii) Mg atom is twice heavier than carbon atom. Justify.

(iii) What is macro molecule? Give example.

(iv) Define partition chromatography with example.

(v) State distribution law.

(vi) How fluted filter paper is prepared?

(vii) State Charle's law. Write its mathematical form.

(viii) Define critical temperature and critical pressure of a substance.

(ix) Differentiate between natural and artificial plasma.

(x) Differentiate between reversible and irreversible reactions.

(xi) Define Buffer capacity.

(xii) What is the effect of common ion on solubility?

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) Why is boiling point of H_2O greater than that of HF?

(ii) What are London forces? Give an example.

(iii) Define lattice energy. Give one example.

(iv) What are molecular solids? What type of interactions hold them together?

(v) Define spectrum. Give its two types.

(vi) The e/m values of positive rays for different gases are different but those for cathode rays, the e/m values are same. Why?

(vii) How are the neutrons involved in the conversion of $^{29}_{65}Cu$ into $^{30}_{66}Zn$?

(viii) What are x-rays? How are they produced?

(ix) Aqueous solution of $CuSO_4$ is acidic in nature. Give the reason.

(x) Why are NaCl and KNO_3 used to lower the melting point of ice?

(xi) What are Pseudo first order reactions? Give one example.

(xii) How does the surface area of reactants affect the rate of reaction? Give an example.

Q4. (Marks 12) Write short answers of any SIX parts.

(i) How does the electronegativity difference decide the nature of ionic bond?

(ii) Why an ionic bond is stronger than covalent bond?

(iii) Why the atomic radii increase down the group?

(iv) How the bond length is affected by hybridization?

(v) What is state and state function?

(vi) What do you mean by internal energy of chemical system?

(vii) Define surroundings and give examples.

(viii) Write the cathodic reaction in fuel cells.

(ix) Give the structure of anode and cathode in lead acid battery.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5. (Marks 1,1,1,1) (a)Write various steps to calculate the empirical formula of a compound.

(b) What is Plank's Quantum Theory? Write its main points.

Q6. (Marks 04) (a)250cm^3 of the sample of hydrogen effuses four times as rapidly as 250cm^3 of an unknown gas. Calculate the molar mass of unknown gas.

(b) Describe fuel cells. Give their uses.

Q7. (Marks 04) (a)Write postulates of M.O.T. and explain oxygen molecule by this theory.

(b) Explain first law of thermodynamics.

Q8. (Marks 04) (a)What are liquid crystals? Give their uses.

(b) The solubility of CaF_2 in water at 25°C is found to be 2.05×10^{-4} mol dm^{-3}. What is the value of K_{sp} at this temperature?

Q9. (Marks 04) (a)Give applications of elevation of boiling point and depression of freezing point.

(b) Explain rate determining step in detail.